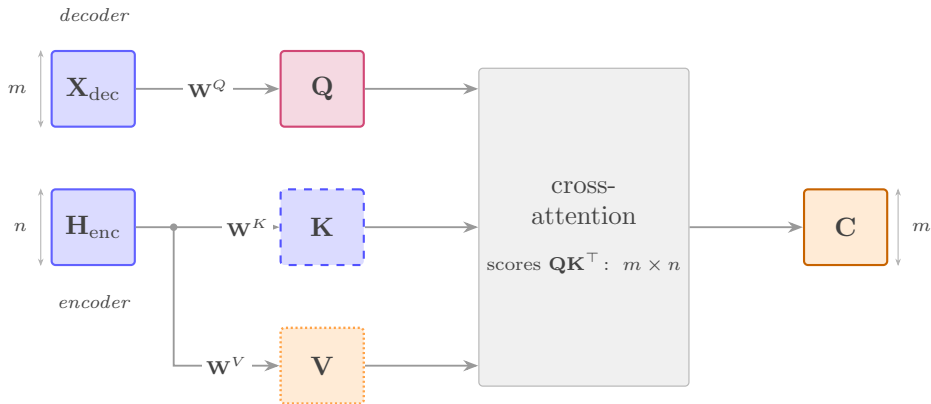


Cross-attention: queries from the decoder, keys/values from the encoder



$$\text{CrossAttn}(\mathbf{X}_{\text{dec}}, \mathbf{H}_{\text{enc}}) = \text{Attn}(\mathbf{X}_{\text{dec}} \mathbf{W}^Q, \mathbf{H}_{\text{enc}} \mathbf{W}^K, \mathbf{H}_{\text{enc}} \mathbf{W}^V)$$

every decoder position attends over all n encoder positions; each output row is a *convex combination* of the encoder values — the decoder selects and averages the encoded source, it cannot fabricate beyond it (example $m=3, n=4 \Rightarrow$ scores 3×4)